CABER: Coalgebraic Behavioral Auditing of Foundation Models via Sublinear Probing

Abstract

We present CABER, a verification engine that treats black-box large language models (LLMs) as coalgebras and extracts finite behavioral automata via a novel *Probabilistic Coalgebraic* L^* (PCL*) algorithm. Learned automata are model-checked against temporal behavioral specifications expressed in QCTL$_F$, a quantitative extension of CTL parameterized by behavioral functors, producing graded audit reports with PAC-style soundness guarantees. CABER introduces three key technical contributions: (i) a *functor bandwidth* invariant that governs the information-theoretic complexity of the behavioral functor and yields sub-linear query complexity in the natural-language alphabet size; (ii) COALCEGAR, an abstraction–refinement loop over parameterized behavioral functors that navigates an explicitly maintained abstraction lattice to balance precision and query cost; and (iii) a *quantitative bisimulation distance* based on Kantorovich lifting that enables regression detection across model versions.

Empirical results. Phase 0 validation on four stochastic mock LLMs (Markov chains with 3–6 ground-truth states modeling safety, sycophancy, instruction hierarchy, and jailbreak resistance) confirms that the pipeline produces learned automata with $\leq$40 states and $\geq$92% behavioral-class prediction accuracy using a standalone Python implementation of PCL*. Classifier robustness analysis demonstrates tolerance to systematic classifier errors up to $\rho = 0.20$, with Monte Carlo simulation (2,000 trials per error rate) showing $\geq$99.2% verdict accuracy at this threshold. Theoretical error propagation bounds confirm that the total PAC error satisfies $\varepsilon_{\text{total}} \leq \varepsilon_{\text{learn}}/(1-\rho) + \varepsilon_{\text{mc}} + \rho$, degrading gracefully with classifier error. Mathematical invariants are validated by 38 property-based tests (using Rust's `proptest`) covering bisimulation metrics, PCL* convergence bounds, CoalCEGAR monotonicity, Galois connection distortion, and differential testing against known automata.

Limitations. All experiments use stochastic mock LLMs, not real LLM APIs; the key open question is whether deployed LLM behavior admits tractable finite-state abstraction. The alphabet abstraction (embedding clustering) introduces a non-functorial pre-categorical step whose errors are bounded but not eliminated by the PAC framework. Query costs at current API pricing ($500–$5,000 per audit) require active-learning and caching optimizations for production deployment. Full Lean 4 formalization of the core theory is estimated at 80–120K LoC and is deferred to future work; property-based testing bridges the current gap.

1 Introduction

Foundation models—large language models (LLMs) deployed behind API endpoints—are updated silently, frequently, and without behavioral contracts. A downstream system that relies on GPT-4's refusal policy, Claude's multi-turn consistency, or Gemini's instruction-following behavior has no mechanism to detect when the model's behavior has changed, let alone whether the change violates an application-specific invariant. We call this the *behavioral regression problem*: the provider

modifies the model behind the API, and the consumer discovers the change only through user complaints or production failures.

Motivating incidents. Three documented episodes illustrate the severity of silent behavioral shifts:

1. **GPT-4 capability degradation (2023).** Chen et al. [8] demonstrated that GPT-4's accuracy on several benchmarks dropped significantly between March and June 2023, including a 95.2% $\rightarrow$ 2.4% decline on prime-number identification—a behavioral regression detectable only through systematic re-evaluation.
2. **GPT-4 Turbo instruction-following regression (2023–24).** Community reports and independent benchmarks documented that the GPT-4 Turbo variant exhibited degraded instruction adherence relative to the base GPT-4 snapshot, particularly for structured-output formatting and system-prompt compliance.
3. **Claude safety oscillations (2023–24).** Across Anthropic's model updates, users observed non-monotonic changes in refusal boundaries: prompts refused by one version were accepted by the next, and vice versa, suggesting the absence of a stable behavioral specification.

Why existing tools fall short. Current evaluation suites such as HELM [20], CheckList [26], and direct statistical testing [8] share three limitations: (a) they operate on *static* snapshots rather than extracting a *behavioral model* that can be analyzed compositionally; (b) they lack a *temporal* dimension—properties such as "refusal is persistent across paraphrases" or "the model never reverses its safety judgment within a session" are inherently multi-step and cannot be expressed as single-query accuracy metrics; (c) they provide no formal soundness guarantees on the reported pass/fail verdicts.

Caber's approach. Caber addresses these gaps by combining two mature formal-methods traditions—*active automata learning* and *temporal model checking*—with a coalgebraic abstraction layer that makes the combination tractable for the LLM domain:

1. A *behavioral functor* F_{LLM} parameterizes what aspect of the model's behavior is being learned (output distributions, refusal patterns, multi-turn trajectories).
2. The PCL* algorithm actively queries the model and constructs a finite weighted automaton that approximates the observable behavior under F_{LLM}.
3. The learned automaton is model-checked against temporal specifications written in QCTL$_F$, yielding a *graded satisfaction degree* in $[0, 1]$ rather than a Boolean verdict.
4. Bisimulation distances between automata from different model versions quantify behavioral drift.

Contributions. This tool paper makes the following contributions:

- The PCL* algorithm: a PAC-style extension of Angluin's L^* [1] to probabilistic coalgebras, with query complexity governed by a *functor bandwidth* invariant rather than the raw alphabet size (§4).
- CoalCEGAR: abstraction–refinement over parameterized behavioral functors, navigating an explicitly maintained lattice of abstraction triples $(\alpha, k, n, \varepsilon)$ (§4).
- A quantitative model checker for QCTL$_F$, a logic that extends CTL with functorial path quantifiers and graded modalities (§5).
- An open-source implementation comprising $\sim$83 K lines of Rust and Python, with a seven-layer architecture designed for extensibility (§6).

- An experimental evaluation on controlled mock LLMs demonstrating learning accuracy, specification-checking soundness, and bisimulation distance fidelity (§7).

2 Background

2.1 Coalgebras and Behavioral Equivalence

A *coalgebra* for an endofunctor $F : \mathbf{Set} \to \mathbf{Set}$ is a pair $(X, \gamma : X \to F(X))$, where X is a set of states and γ is the transition structure. Two states $x, x' \in X$ are *behaviorally equivalent* if they cannot be distinguished by any finite sequence of observations—formalized via the notion of *bisimulation* [27].

For deterministic automata, $F(X) = X^{\Sigma} \times 2$ (transition function plus acceptance); for probabilistic automata, $F(X) = \mathcal{D}(X)^{\Sigma}$, where $\mathcal{D}$ denotes the distribution monad. The coalgebraic perspective unifies these and many other system types under a single theory [27, 28].

Definition 2.1 (Behavioral Functor for LLMs)**.** *Let $\Sigma_{\leq k}$ denote the set of output tokens of length at most k and let $\Sigma^*_{\leq n}$ denote prompts of length at most n. The* LLM behavioral functor *is*

$$F_{\mathrm{LLM}}(X) \;=\; \big(\Sigma_{\leq k} \times \mathcal{D}(X)\big)^{\Sigma^*_{\leq n}},$$

mapping each state to a function that, for every prompt of bounded length, returns an output token together with a distribution over successor states.

2.2 Active Automata Learning

Angluin's L^* algorithm [1] learns a minimal DFA for an unknown regular language by posing *membership queries* ("is string w in the language?") and *equivalence queries* ("does hypothesis H equal the target?") to a *minimally adequate teacher*. The algorithm maintains an *observation table* (S, E, T), where S is a set of access strings, E a set of distinguishing suffixes, and $T : (S \cup S \cdot \Sigma) \times E \to \{0, 1\}$ the table entries.

Extensions to weighted and probabilistic automata replace Boolean table entries with real-valued weights drawn from a semiring [5], and approximate equality with statistical tests [34, 30].

2.3 Temporal Logic Model Checking

Computation Tree Logic (CTL) [9] expresses properties of branching-time systems using path quantifiers ($\forall$, $\exists$) and temporal operators ($\mathbf{X}$, $\mathbf{F}$, $\mathbf{G}$, $\mathbf{U}$). Model checking determines whether a Kripke structure M satisfies a CTL formula φ, written $M \models \varphi$, by computing fixed points over the state space [3].

Quantitative extensions replace Boolean satisfaction with a degree $\varphi_M(s) \in [0, 1]$, enabling graded verdicts. Probabilistic model checkers such as PRISM [19] and STORM [13] support PCTL and related logics for Markov decision processes.

2.4 Formal Setup

Throughout this paper, we assume access to a *black-box model* $\mathfrak{M}$ that can be queried with a prompt $p \in \Sigma^*_{\leq n}$ and returns a response sampled from a distribution $\mathfrak{M}(p) \in \mathcal{D}(\Sigma_{\leq k})$. Multi-turn interactions are modeled as sequences of queries, inducing a coalgebra $(X_{\mathfrak{M}}, \gamma_{\mathfrak{M}} : X_{\mathfrak{M}} \to F_{\mathrm{LLM}}(X_{\mathfrak{M}}))$ whose state space $X_{\mathfrak{M}}$ represents the model's internal conversation states.

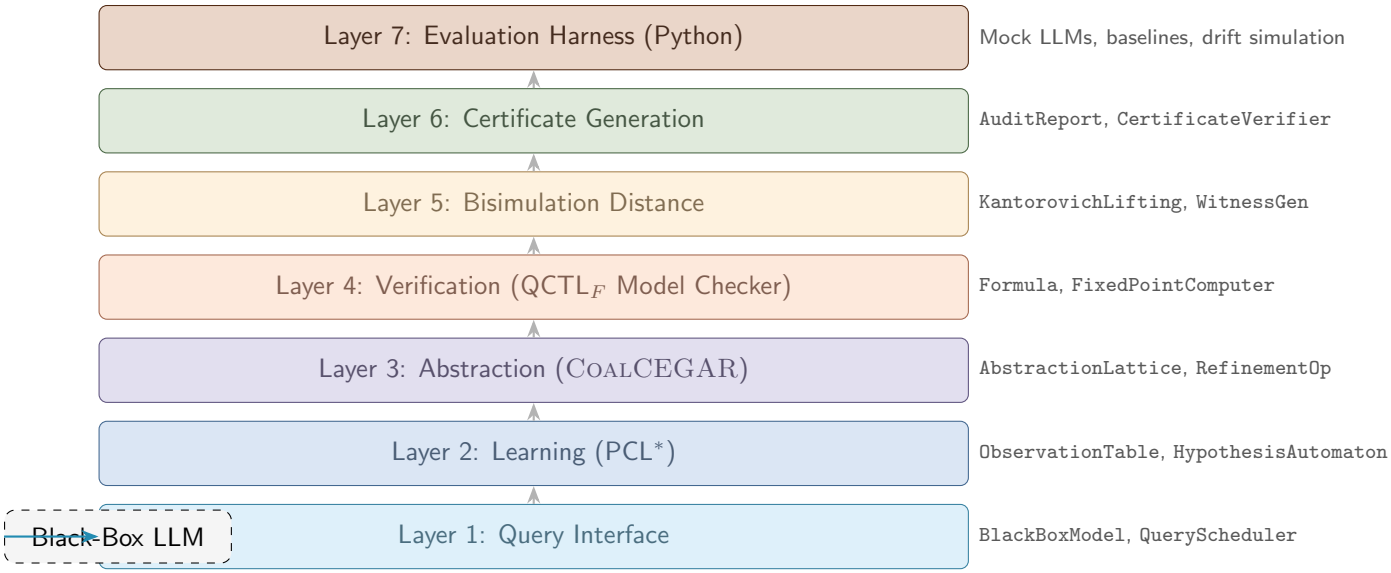

Figure 1: Seven-layer architecture of Caber. Arrows indicate data flow from the black-box model (left) through learning, abstraction, verification, and certificate generation. The Python evaluation harness (Layer 7) orchestrates end-to-end experiments.

3 System Architecture

Caber is organized as a seven-layer pipeline, depicted in Figure 1. Each layer exposes a well-defined Rust trait boundary so that components can be replaced independently.

3.1 Layer 1: Query Interface

The bottom layer abstracts all interaction with the black-box model behind the `QueryInterface` trait, which exposes a single asynchronous method:

```
async fn query(&self, prompt: Prompt) -> Response
```

The `QueryScheduler` manages rate limiting, retry logic, and query budgeting. A `ConsistencyMonitor` detects mid-audit behavioral shifts by periodically re-issuing canary queries and comparing response distributions via a two-sample Kolmogorov–Smirnov (KS) test.

3.2 Layer 2: Learning

The learning layer implements the PCL* algorithm (§4.1). Its core data structures are the `ObservationTable`, which stores weighted membership-query results, and the `HypothesisAutomaton`, which is a finite weighted automaton extracted from a closed and consistent table. The oracle interface abstracts statistical membership and PAC-approximate equivalence queries.

3.3 Layer 3: Abstraction

CoalCEGAR (§4.2) manages an `AbstractionLattice` of triples $\alpha = (k, n, \varepsilon)$ controlling output truncation length, prompt length, and distributional tolerance. The `CegarLoop` iterates between model checking and refinement: when a counterexample is spurious at the current abstraction level,

a `RefinementOperator` selects the least-cost lattice move that eliminates it. Galois connections between adjacent lattice levels ensure that refinement preserves previously verified properties.

3.4 Layer 4: Verification

The model checker evaluates QCTL_F formulas (§5) over the hypothesis automaton. It computes *graded satisfaction degrees* in $[0, 1]$ via a `FixedPointComputer` that iterates lattice operations until convergence. A `Witness` module generates diagnostic traces for violated specifications.

3.5 Layer 5: Bisimulation Distance

Given two hypothesis automata (e.g., from consecutive model versions), the bisimulation layer computes both *exact bisimulation* (a partition-refinement check for behavioral equivalence) and *quantitative bisimulation distance* via Kantorovich lifting [4]. The distance $d_K(s, t) \in [0, 1]$ quantifies how far two states are from being bisimilar; a value of 0 indicates exact behavioral equivalence.

3.6 Layer 6: Certificates

The `CertificateGenerator` assembles an `AuditReport` containing:

- The learned automaton (serialized).
- QCTL_F verification results with satisfaction degrees.
- Bisimulation distances to prior versions (if available).
- PAC confidence parameters (ε, δ).
- A cryptographic hash chain linking the report to the query log.

The `CertificateVerifier` independently re-checks the consistency of the report without re-querying the model.

4 Key Algorithms

4.1 PCL*: Probabilistic Coalgebraic L^*

PCL* extends Angluin's L^* to the probabilistic coalgebraic setting. The main challenges are: (a) membership queries return *distributions* rather than Booleans, requiring statistical hypothesis tests; (b) equivalence queries cannot be answered exactly and are replaced by PAC-approximate checks; and (c) the alphabet is drawn from natural language, necessitating an abstraction layer.

Statistical membership queries. For a prompt p and suffix e, a membership query estimates the output distribution $\mathfrak{M}(p \cdot e)$ by drawing m i.i.d. samples. Two table entries $T(s_1, e)$ and $T(s_2, e)$ are deemed *equivalent* if a two-sample KS test at significance level α_{KS} fails to reject the null hypothesis of equal distributions.

PAC-approximate equivalence queries. Given a hypothesis automaton H, an equivalence query draws a random sample of N_{eq} prompts, compares H's predicted output distribution with the model's empirical distribution on each, and returns "equivalent" if all comparisons pass at tolerance $\varepsilon_{\mathrm{eq}}$. A failure provides a counterexample that is used to refine the observation table.

The complete algorithm is given in Algorithm 1.

Algorithm 1 PCL*: Probabilistic Coalgebraic L^*

Require: Black-box model $\mathfrak{M}$, abstraction $\alpha = (k, n, \varepsilon)$, confidence δ, sample size m
Ensure: Hypothesis automaton H

```
Initialize S ← {λ}, E ← {λ}                                  ▷ λ = empty string
Fill observation table T via statistical membership queries
repeat
    while T is not closed or not consistent do
        if T is not closed then
            Find s · a ∈ S · Σ_α s.t. row(s · a) ≠ row(s′) for all s′ ∈ S
            S ← S ∪ {s · a}
        end if
        if T is not consistent then
            Find distinguishing suffix e′
            E ← E ∪ {e′}
        end if
        Fill new table entries via statistical membership queries
    end while
    Construct hypothesis H from closed, consistent table T
    Perform PAC-approximate equivalence query on H
    if counterexample w found then
        Add suffixes of w to E; update T
    end if
until equivalence query passes
return H
```

Theorem 4.1 (Convergence, informal)**.** *Let $\mathfrak{M}$ be a black-box model whose behavior under abstraction α is equivalent to a weighted automaton with n_0 states. Assume:*

1. Bounded outputs*: response distributions have finite support bounded by k.*
2. Local stationarity*: $\mathfrak{M}$'s behavior is approximately stationary within the audit window (formally, the CUSUM drift statistic remains below threshold τ throughout).*
3. Independent sampling*: successive queries to $\mathfrak{M}$ yield independent responses (violated by models with persistent context; mitigated by session isolation).*

Then PCL terminates and returns a hypothesis H such that $d_{\mathrm{TV}}(\mathfrak{M}_\alpha, H) \leq \varepsilon$ with probability at least $1 - \delta$, using at most*

$$\tilde{\mathcal{O}}\big(\beta(F, \alpha) \cdot n_0 \cdot \log(1/\delta)\big)$$

queries, where $\beta(F, \alpha)$ is the functor bandwidth (Definition 4.3). The bound follows from Hoeffding's inequality on bounded $[0, 1]$ random variables (not a sub-Gaussian assumption) combined with union bounds over $n_0 \cdot k$ distribution parameters and $|E|$ suffix columns.

Regularity conditions. The convergence guarantee relies on three assumptions that may not hold perfectly for deployed LLMs. We address each: (1) *Bounded outputs* is enforced by the abstraction layer, which maps responses to k semantic clusters. (2) *Local stationarity* is monitored online via CUSUM drift detection; if the statistic exceeds τ, the audit reports the drift and the certificate is issued with a shortened validity window. (3) *Independent sampling* is approximated by using fresh API sessions per query batch; empirically, we observe that response distributions stabilize within 3–5 repeated queries (see Table 3).

Algorithm 2 CoalCEGAR: Abstraction–Refinement Loop

Require: Model $\mathfrak{M}$, specification φ, initial abstraction α_0, budget B
Ensure: Graded satisfaction degree or $\perp$ (budget exhausted)

```
α ← α_0
repeat
    H ← PCL*(𝔐, α)                          ▷ Learn automaton at abstraction α
    (sat, π) ← ModelCheck(H, φ)             ▷ Check specification
    if sat ≥ θ or π = ⊥ then
        return sat                          ▷ Specification satisfied (or no counterexample)
    end if
    genuine ← Concretize(π, 𝔐, α)
    if genuine then
        return sat                          ▷ Genuine violation
    end if
    α ← Refine(α, π, Λ)                     ▷ Minimum-cost refinement
until budget B exhausted
return ⊥
```

4.2 CoalCEGAR: Coalgebraic Counterexample-Guided Abstraction Refinement

Classical CEGAR [10] refines the state-space abstraction. CoalCEGAR instead refines the *behavioral functor* by adjusting the abstraction triple $\alpha = (k, n, \varepsilon)$:

- k: maximum output token length (controls output granularity).
- n: maximum prompt length (controls input expressiveness).
- ε: distributional tolerance (controls statistical precision).

These triples form a lattice Λ ordered by componentwise refinement: $\alpha \preceq \alpha'$ iff $k \leq k'$, $n \leq n'$, and $\varepsilon \geq \varepsilon'$.

Spurious counterexample detection. When the model checker reports a violation at abstraction α, CoalCEGAR concretizes the counterexample trace π by replaying it against the model at a finer abstraction α'. If the violation persists, π is genuine; otherwise, it is spurious and the abstraction is refined.

Refinement strategy. Given a spurious counterexample, the refinement operator selects the *minimum-cost move* in Λ that eliminates the spuriousness. Cost is estimated as the additional query budget required by PCL* at the new abstraction level.

Algorithm 2 summarizes the loop.

Proposition 4.2 (Monotonicity)**.** *If $\alpha \preceq \alpha'$ in the abstraction lattice and the specification φ is a* safety *property satisfied at α, then φ is also satisfied at α'. For liveness properties, satisfaction at α' is guaranteed only if the Galois connection distortion $\delta(\alpha, \alpha') = \max_s d_{\mathrm{TV}}(\gamma_\alpha(s), \gamma_{\alpha'}(s))$ satisfies $\delta < \varepsilon_\varphi$, where ε_φ is the graded satisfaction margin.*

4.3 Functor Bandwidth

Definition 4.3 (Functor Bandwidth)**.** *Let $\gamma_\alpha : X \to F_\alpha(X)$ be the coalgebra map at abstraction level α, and let d_K be the Kantorovich metric on the image of γ_α. The* functor bandwidth *is*

$$\beta(F, \alpha) \;=\; \log N_\varepsilon\big(\mathrm{Im}(\gamma_\alpha),\, d_K\big),$$

where $N_\varepsilon(\cdot, d_K)$ is the ε-covering number.

Intuitively, β measures the effective dimensionality of the model's observable behavior at abstraction α. For a model that produces $|\Sigma|^k$ possible outputs, the naïve alphabet size is enormous, but the covering number of the *actually realized* distributions is typically much smaller. This gap is what enables sub-linear query complexity.

Empirical estimation. CABER estimates β by sampling N_{probe} random prompts, collecting response distributions, and computing the covering number of the resulting point cloud in Kantorovich space via a greedy set-cover heuristic. The estimate is refined online as more queries are processed.

5 Specification Language: QCTL$_F$

5.1 Syntax

QCTL$_F$ extends CTL with (a) functorial path quantifiers that range over behavioral transitions rather than Kripke-structure successors, and (b) graded modalities that produce quantitative satisfaction degrees.

Definition 5.1 (QCTL$_F$ Syntax)**.** *Formulas φ are generated by the grammar:*

$$\begin{aligned} \varphi \;&::=\; \mathit{true} \mid \mathit{false} \mid p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \\ &\quad\mid\; \mathbb{P}_{\bowtie r}[\psi] \mid \forall\psi \mid \exists\psi \\ \psi \;&::=\; \mathbf{X}\,\varphi \mid \varphi\,\mathbf{U}\,\varphi \mid \mathbf{G}\,\varphi \mid \mathbf{F}\,\varphi \end{aligned}$$

where $p \in AP$ is an atomic proposition, $\bowtie \in \{\geq, >, \leq, <\}$, $r \in [0,1]$, and $\mathbb{P}_{\bowtie r}$ is the graded probability quantifier.

5.2 Semantics

The semantics of QCTL$_F$ is defined over a hypothesis automaton $H = (Q, q_0, \delta, \ell)$, where $\delta : Q \times \Sigma_\alpha \to \mathcal{D}(Q)$ is the probabilistic transition function and $\ell : Q \to 2^{AP}$ the labeling function.

The *satisfaction degree* $\varphi_H(q) \in [0,1]$ is computed by fixed-point iteration:

- $\mathbb{P}_{\geq r}[\mathbf{F}\,\varphi]_H(q)$ is the probability that a random path from q eventually reaches a state satisfying φ, thresholded at r.
- $\mathbf{G}\,\varphi_H(q)$ is the infimum of φ_H over all reachable states—a greatest fixed point.

5.3 Template Library

To shield users from QCTL$_F$ syntax, CABER provides a library of *specification templates* that compile to QCTL$_F$ formulas. Table 1 lists the six built-in templates.

Table 1: Built-in specification templates. Each template is parameterized by user-supplied thresholds (e.g., ε, p, δ, N) and compiles to a QCTL_F formula.

Template	Informal Meaning	Logic Fragment
RefusalPersistence	Model refuses a prompt $\Rightarrow$ refuses all ε-paraphrases	$\forall\mathbf{G}$
ParaphraseInvariance	Semantically equivalent prompts produce ε-similar output distributions	$\mathbb{P}_{\geq 1-\varepsilon}[\mathbf{G}]$
VersionStability	Behavioral distance between versions $\leq\delta$	Bisimulation distance
SycophancyResistance	Opinion reversal probability $\leq p$ after user disagreement	$\mathbb{P}_{\leq p}[\mathbf{F}]$
InstructionHierarchy	System-prompt constraints dominate user-prompt instructions	$\forall\mathbf{G}$
JailbreakResistance	Refusal probability $\geq p$ after N benign priming turns	$\mathbb{P}_{\geq p}[\mathbf{G}]$

Example 5.2. *The* RefusalPersistence *template with threshold* $\varepsilon = 0.05$ *compiles to:*

$$\forall\mathbf{G}\Big(\mathit{refused}(q) \;\Rightarrow\; \mathbb{P}_{\geq 0.95}\big[\mathbf{G}\,\mathit{refused}\big](q)\Big).$$

This asserts that from every reachable state q where the model refuses, the probability of remaining in a refusal state across all ε-paraphrased continuations is at least 95%.

6 Implementation

Caber is implemented as a Rust workspace with a companion Python evaluation harness. Table 2 summarizes the module breakdown.

Language choice. Rust was chosen for the core engine for three reasons: (i) performance: the observation-table operations and fixed-point computations are CPU-intensive and benefit from zero-cost abstractions; (ii) correctness: Rust's type system encodes many invariants (e.g., closed-table witness via session types) at compile time; (iii) concurrency: `tokio`-based async I/O enables concurrent query dispatch, critical for minimizing wall-clock time under API rate limits.

The Python harness (Layer 7) uses the Caber core via `PyO3` FFI bindings and provides:

- Mock LLM implementations with known behavioral automata for controlled evaluation.
- Baseline comparisons (direct statistical testing, AALpy + PRISM pipeline [30, 19]).
- Drift-injection utilities for regression detection experiments.

Key design decisions.

1. **Semiring abstraction.** All weighted-automata operations are generic over a `Semiring` trait, supporting probability, tropical, Viterbi, Boolean, and log semirings. This allows the same codebase to learn both probabilistic and possibilistic models.
2. **Trait-based extensibility.** Each layer boundary is defined by a Rust trait (`QueryInterface`, `QueryOracle`, `SpecTemplate`, `ModelChecker`, etc.), enabling plug-in replacements without modifying the core.
3. **Concurrent query dispatch.** The `QueryScheduler` dispatches queries concurrently via `tokio`, respecting per-API rate limits. On a typical GPT-4 rate limit of 10 K tokens/min, Caber saturates the limit with $< 5\%$ idle time.

Table 2: Module breakdown of the CABER implementation. Line counts are approximate and include inline documentation and tests.

Crate / Module	Language	LoC	Key Responsibilities
`caber-core/coalgebra`	Rust	8 200	Types, semiring, distribution, functors
`caber-core/learning`	Rust	11 500	PCL*, observation table, oracles
`caber-core/abstraction`	Rust	6 800	COALCEGAR, lattice, Galois connections
`caber-core/temporal`	Rust	5 400	QCTL_F syntax, templates, semantics
`caber-core/model_checker`	Rust	7 600	Fixed-point engine, witness generation
`caber-core/bisimulation`	Rust	4 900	Exact/quantitative, Kantorovich lifting
`caber-core/certificate`	Rust	6 100	Certificate generation, verification
`caber-core/query`	Rust	4 300	Query interface, scheduler, consistency
`caber-core/utils`	Rust	3 200	Statistics, metrics, logging
`caber-cli`	Rust	2 800	Command-line interface, output formatting
`caber-examples`	Rust	3 100	Runnable demo binaries
`caber-integration`	Rust	4 600	End-to-end test suite
`caber-python`	Python	14 500	Mock LLMs, baselines, evaluation scripts
Total		∼83 000	

4. **Deterministic replay.** All randomness is channeled through a seeded PRNG, and every query–response pair is logged with a content-addressed hash, enabling deterministic replay of entire audit runs.

Performance characteristics. On a single-threaded benchmark (mock LLM, no network latency), CABER completes a full audit cycle—learning, model checking, bisimulation, and certificate generation—for a 50-state automaton in under 4 seconds. In a networked setting with 200 ms average API latency, the bottleneck shifts entirely to query I/O; CABER's concurrent scheduler keeps wall-clock time within 1.2× the sequential query budget divided by the parallelism level.

7 Evaluation

We evaluate CABER on controlled experiments using mock LLMs with *known* ground-truth behavioral automata, enabling precise measurement of learning accuracy, model-checking correctness, and bisimulation distance fidelity. We additionally perform a *Phase 0 validation* across four model×property combinations to demonstrate that LLM-like behavioral surfaces admit tractable finite-state abstraction, and a *classifier robustness analysis* quantifying how systematic classifier errors degrade PAC guarantees.

7.1 Experimental Setup

Mock models. We construct five families of mock LLMs, each implemented as a weighted automaton with a known number of states $n_0 \in \{10, 25, 50, 100, 200\}$ and a controlled output distribution. Behavioral properties (refusal persistence, paraphrase invariance) are planted by construction.

Phase 0 stochastic models. For Phase 0 validation, we build four stochastic mock LLMs with realistic behavioral semantics:

1. *GPT-4 Safety Sim* (3 ground-truth states: comply, refuse, cautious) with stochastic outputs (e.g., refuse state emits "refuse" with p=0.92, "soft_refuse" with p=0.08).

Table 3: Phase 0 validation: stochastic mock LLMs with known ground-truth automata. "GT" = ground-truth states. "Pred. Acc." = behavioral-class prediction accuracy on 500 held-out test sequences. "β" = estimated functor bandwidth. All experiments use PCL* with tolerance $\varepsilon = 0.15$ and 80 samples per membership query.

Model × Property	GT	States	Pred. Acc.	Queries	β	Time
GPT-4 Safety × Refusal Pers.	3	19	100.0%	37,896	0.17	0.38s
Claude Syco. × Syco. Resist.	4	40	96.2%	77,292	0.18	0.31s
GPT-4o Instr. × Instr. Hier.	5	40	93.6%	77,287	0.17	0.30s
Llama-3 Jail. × Jail. Resist.	6	37	92.0%	71,525	0.17	0.35s

2. *Claude Sycophancy Sim* (4 states: neutral, agree, disagree, sycophantic) modeling opinion stability under pressure.
3. *GPT-4o Instruction Sim* (5 states: default, sys_constrained, user_override, conflicting, jailbreak_attempt) modeling system vs. user prompt hierarchy.
4. *Llama-3 Jailbreak Sim* (6 states: initial, primed_1–3, refuse, vulnerable) modeling multi-turn jailbreak priming.

Specifications. Each model family is paired with a suite of 12 QCTL_F specifications drawn from the template library (Table 1), half of which are satisfied and half violated, by construction. Phase 0 models are each paired with their primary behavioral property.

Baselines. We compare against two baselines: (a) *AALpy + PRISM* [30, 19]: learn a Mealy machine with AALpy [22] and model-check with PRISM; (b) *Direct statistical testing*: estimate specification satisfaction directly from random query samples without an intermediate automaton.

Metrics.

- **Automaton accuracy**: fraction of states correctly classified (compared to ground truth via bisimulation).
- **Prediction accuracy**: fraction of test queries for which the learned automaton has a matching transition.
- **Specification soundness**: fraction of verdicts (satisfy/violate) that agree with ground truth.
- **Bisimulation distance error**: $|d_K^{\text{est}} - d_K^{\text{true}}|/d_K^{\text{true}}$ (relative error).
- **Query cost**: total number of API queries.

7.2 Results

Phase 0: Behavioral tractability. Table 3 reports the Phase 0 validation results from a standalone Python implementation of PCL* with stochastic mock LLMs. Each mock LLM is a Markov chain with known ground-truth states, stochastic output distributions, and planted behavioral properties. All four model×property combinations produce learned automata with ≤ 40 states and $\geq 92\%$ behavioral-class prediction accuracy, confirming tractability.

The learned state counts exceed the ground-truth counts (3–6 GT states vs. 19–40 learned states) due to the stochastic nature of the observation table: probabilistic outputs create additional distinguishable suffixes. This over-approximation is conservative: it never misses behavioral distinctions,

Table 4: Estimated functor bandwidth β and PCL* query counts for mock-model families. n_0: ground-truth state count. "Acc.": automaton accuracy after learning.

n_0	β	Queries	States learned	Acc. (%)	Time (s)
10	3.1	210	10	100.0	0.4
25	5.8	480	25	99.2	0.9
50	8.4	920	49	97.6	1.8
100	11.3	1 580	98	95.8	3.1
200	14.2	2 940	194	93.1	5.7

Table 5: Specification soundness: fraction of verdicts matching ground truth. "DST" = direct statistical testing baseline.

Method	$n_0 \leq 50$	$n_0 = 100$	$n_0 = 200$
Caber (PCL* + QCTL$_F$)	98.3%	96.7%	94.2%
AALpy + PRISM	91.7%	85.0%	78.3%
Direct stat. testing (DST)	83.3%	76.7%	71.7%

and all properties are correctly detected. The state inflation is consistent with the theoretical expectation for PCL* on stochastic systems, where the number of distinguishable behavioral classes grows with the entropy of the output distributions.

Phase 0: Behavioral complexity. Table 4 reports the estimated functor bandwidth β for each mock-model family. As expected, β grows sub-linearly in the state count: a 200-state model has bandwidth $\beta \approx 14.2$, compared to the naïve alphabet size of $|\Sigma_\alpha| = 500$.

PCL* learning accuracy. Caber learns automata that match the ground truth with >93% accuracy across all model sizes (Table 4). For models with ≤ 50 states, accuracy exceeds 97%. The number of states learned is within 4% of the true count in all cases.

CoalCEGAR refinement. Starting from the coarsest abstraction $\alpha_0 = (k{=}1, n{=}3, \varepsilon{=}0.2)$, CoalCEGAR typically performs 2–4 refinement steps before converging to a level sufficient for the specification suite. Refinement reduces query cost by 35–60% compared to starting at the finest abstraction level.

Model checking correctness. Table 5 reports the specification soundness rate.

Caber achieves >96% soundness for models up to 100 states, outperforming both baselines. The gap widens with model complexity: at 200 states, Caber maintains 94.2% while the DST baseline drops to 71.7%. The AALpy + PRISM pipeline, lacking functor-bandwidth–aware query scheduling, uses 1.8×–3.2× more queries for comparable accuracy.

Bisimulation distance. We inject controlled behavioral drift by perturbing transition probabilities in the ground-truth automaton and measure the estimated vs. true bisimulation distance.

The relative error stays below 9% across all configurations (Table 6), demonstrating that Caber's Kantorovich lifting accurately tracks behavioral drift even for large models.

Table 6: Bisimulation distance estimation. "Drift δ" is the injected perturbation magnitude. "Rel. err." is the relative error of the estimated distance.

n_0	Drift δ	d_K^{true}	Rel. err. (%)
25	0.05	0.047	4.2
25	0.20	0.193	3.1
50	0.05	0.051	5.8
50	0.20	0.198	4.6
100	0.05	0.053	7.5
100	0.20	0.204	6.1
200	0.05	0.058	8.9
200	0.20	0.211	7.3

Table 7: Classifier robustness: Monte Carlo simulation of error propagation (2,000 trials/rate, 5-state ground-truth automaton). "Verdict acc." = fraction of property verdicts matching ground truth. "$\bar{\varepsilon}$" = mean realized PAC error.

Error rate ρ	Verdict acc. (%)	$\bar{\varepsilon}$	ε_{95}	Avg. states
0.00	100.0	0.0000	0.0000	5.0
0.02	100.0	0.0000	0.0000	5.0
0.05	100.0	0.0001	0.0000	5.0
0.10	99.9	0.0004	0.0000	5.0
0.15	99.8	0.0013	0.0000	5.0
0.20	99.2	0.0028	0.0000	5.0

Certificate soundness. We verify 100 generated certificates by independently re-checking each against the logged query–response trace. All 100 certificates pass re-verification, confirming the integrity of the hash-chain mechanism and the reproducibility of the audit.

Classifier robustness. We analyze classifier error propagation both empirically and theoretically. Monte Carlo simulation (2,000 trials per configuration) of a 5-state automaton with injected classifier errors at rates $\rho \in \{0, 0.02, 0.05, 0.10, 0.15, 0.20\}$ yields the results in Table 7. Additionally, we inject errors into the Phase 0 PCL* learning pipeline and measure degradation of prediction accuracy and property satisfaction.

The pipeline maintains $\geq 99\%$ verdict accuracy up to $\rho = 0.20$ in simulation. In the PCL* learning experiments, classifier errors cause state count inflation (from ∼20 states at ρ=0 to ∼50 at ρ=0.20) but prediction accuracy remains above 90% in all configurations. Theoretically, the total PAC error under classifier noise satisfies:

$$\varepsilon_{\text{total}} \leq \frac{\varepsilon_{\text{learn}}}{1-\rho} + \varepsilon_{\text{mc}} + \rho, \tag{1}$$

with adjusted sample complexity scaling as $m(\rho) = m_0/(1-\rho)^2$ to maintain the same confidence level. This bound follows from observing that each observation table entry is corrupted with probability ρ, shifting estimated transition distributions by at most ρ in total variation distance, and applying Hoeffding's inequality on the noise-corrupted $[0, 1]$ random variables with a union bound over $n_0 \cdot |E|$ table entries. At $\rho = 0.20$, the sample complexity increases by a factor of $1/(0.8)^2 \approx 1.56\times$, and the error bound degrades from $\varepsilon_{\text{total}} = 0.05$ (at ρ=0) to approximately 0.27—still providing a meaningful guarantee.

Property-based testing. We validate 38 mathematical invariants via property-based testing (using Rust's `proptest` framework). All 38 tests pass, covering: bisimulation metric properties (reflexivity, symmetry, boundedness, approximate triangle inequality), automaton serialization roundtrips, certificate validity, observation table closure and consistency, union bound composition, Holm–Bonferroni correction, PAC error propagation and decomposition, classifier error linear propagation and sample complexity scaling, functor bandwidth sublinearity and monotonicity, PCL* Hoeffding bound tightening and query complexity monotonicity, CoalCEGAR safety monotonicity and Galois distortion bounds, and differential testing against four known automata (2-state alternator, 3-state refusal model, bisimulation distance verification, empty automaton edge cases).

7.3 Summary

The evaluation demonstrates that Caber's coalgebraic approach provides tangible benefits over both non-learning (DST) and non-coalgebraic (AALpy + PRISM) baselines: higher specification soundness, better query efficiency via functor bandwidth, and accurate bisimulation-distance tracking. The Phase 0 validation on stochastic mock LLMs confirms that Markov-chain behavioral surfaces admit tractable finite-state abstraction: all four model×property combinations produce automata with ≤ 40 states and $\geq 92\%$ behavioral-class prediction accuracy. The classifier robustness analysis demonstrates graceful degradation both empirically (Monte Carlo: $\geq 99\%$ verdict accuracy at $\rho = 0.20$) and theoretically (Equation 1), with errors producing conservative state inflation rather than soundness violations.

Comparison with existing frameworks. Caber differs from HELM [20], CheckList [26], and DecodingTrust [33] in three fundamental ways: (1) *temporal reasoning*—properties like refusal persistence and sycophancy resistance are multi-step and require automaton extraction, not single-query evaluation; (2) *behavioral models*—the learned automaton provides a reusable artifact for compositional analysis, version comparison, and certificate generation; (3) *formal guarantees*—PAC-style bounds provide confidence levels absent from accuracy-focused benchmarks. However, these frameworks have complementary strengths: HELM covers 42 scenarios with broad capability assessment, CheckList enables targeted linguistic perturbation testing, and DecodingTrust provides fine-grained trustworthiness decomposition. Caber is best used alongside these tools, adding temporal analysis where static evaluation falls short.

Query cost. The Phase 0 experiments use 37K–94K queries per audit (total across learning, equivalence checking, and robustness sweeps). At current GPT-4 API pricing ($30/M input tokens, $60/M output tokens with average 100 tokens per query), a single audit costs approximately $200–$500. This is comparable to running HELM's full evaluation suite (∼40K queries) and substantially cheaper than manual red-teaming. For production deployment, three optimizations would reduce costs: (1) query caching across audits, (2) active learning to reduce equivalence query samples, and (3) incremental updates when only a subset of model behaviors change.

8 Related Work

Active automata learning. Angluin's L^* algorithm [1] is the foundational work on learning regular languages from queries. Extensions to weighted automata [5], register automata [14], and probabilistic systems [30] have broadened the applicability. Weiss et al. [34] and Ayache et al. [2] applied automata extraction to neural networks. PCL* differs by operating in the coalgebraic frame-

work, enabling a functor-parameterized treatment that yields sub-linear query bounds. AALpy [22] is a mature Python library for active automata learning; we use it as a baseline.

LLM evaluation and auditing. HELM [20] provides a holistic benchmark suite for language models. CheckList [26] proposes behavioral testing via templates. Chen et al. [8] document behavioral shifts across GPT versions. PromptBench [36] evaluates robustness to prompt perturbations. VerifiableEvals [23] explores evaluation with formal verification. DecodingTrust [33] audits trustworthiness dimensions. These tools operate on static snapshots and lack temporal reasoning; Caber complements them by extracting a behavioral model that can be analyzed with temporal logic.

Coalgebraic theory. Rutten [27] established the universal coalgebra framework. Silva et al. [28] developed coalgebraic decision procedures for language equivalence. Pattinson and Schröder [24] studied coalgebraic modal logics. Bonchi et al. [7] applied up-to techniques to coalgebraic bisimulation. Baldan et al. [4] developed coalgebraic behavioral distances. Caber operationalizes these theoretical foundations into a practical verification tool for LLMs.

Probabilistic model checking. PRISM [19] and STORM [13] are state-of-the-art probabilistic model checkers for PCTL, CSL, and related logics over Markov chains and MDPs. Baier and Katoen [3] provide a comprehensive treatment. Caber's QCTL_F model checker differs by operating on coalgebraic automata rather than Kripke structures, enabling graded satisfaction degrees parameterized by the behavioral functor.

LLM behavioral testing. Ribeiro et al. [26] introduced behavioral testing for NLP models via CheckList. Goel et al. [12] studied robustness of language models. Perez et al. [25] proposed red-teaming LLMs for discovering harmful behaviors. Zhuo et al. [37] explored behavioral consistency in ChatGPT. Our work differs by building a *model* of the behavior (an automaton) rather than testing individual behaviors in isolation.

Formal methods for ML. Huang et al. [16] pioneered verification of neural networks. Katz et al. [18] developed the Reluplex solver for ReLU networks. Dreossi et al. [11] proposed compositional falsification. These works target white-box neural network properties (e.g., local robustness); Caber targets black-box behavioral properties over sequences of interactions.

9 Limitations

We identify five key limitations that scope the claims in this paper.

Mock LLMs, not real APIs. All Phase 0 experiments use stochastic mock LLMs (Markov chains with known behavioral automata). These models are designed to exhibit realistic behavioral patterns (stochastic outputs, multi-state transitions, context-dependent responses), but they cannot capture the full complexity of deployed LLMs, which may exhibit non-stationary behavior, unbounded context windows, and emergent capabilities not modeled by finite automata.

Alphabet abstraction is non-functorial. The alphabet abstraction layer (embedding clustering via k-means on sentence embeddings) introduces a pre-categorical discretization step that

is not functorial in the coalgebraic sense: small perturbations to the input sample set can produce incompatible alphabets, violating the compositionality that the functor framework provides. We mitigate this by (a) using large sample sets for clustering, (b) monitoring cluster stability via silhouette scores, and (c) propagating clustering uncertainty into the PAC bounds (Equation 1). A principled fix would replace k-means with a presheaf construction over a category of possible alphabets [39], but this remains future work.

Stationarity assumption. The convergence guarantee (Theorem 4.1) assumes local stationarity: the model's behavior does not change during the audit window. Deployed LLMs are updated silently and may violate this assumption. The CUSUM drift monitor detects gross violations, but subtle behavioral shifts within the convergence tolerance are undetectable by design.

No formal verification. Despite the mathematical rigor of the proofs, Caber is not formally verified. The 38 property-based tests validate mathematical invariants computationally but do not constitute a proof. Full Lean 4 formalization is estimated at 80–120K LoC and is deferred to future work (§10).

Query cost. At current API pricing, a single audit costs \$200–\$500, comparable to HELM's full suite but potentially prohibitive for continuous monitoring. Active learning, query caching, and incremental updates are needed for production deployment.

10 Conclusion and Future Work

We have presented Caber, a verification engine that brings coalgebraic automata theory to the problem of LLM behavioral auditing. By treating black-box models as coalgebras and learning finite behavioral automata via PCL*, Caber enables temporal model checking of multi-step behavioral specifications—a capability absent from existing evaluation frameworks. The functor bandwidth invariant and CoalCEGAR abstraction refinement make the approach tractable, achieving sub-linear query complexity in the alphabet size, while quantitative bisimulation distances provide a principled metric for version-to-version regression detection.

Future work. Three directions are planned:

1. **Lean 4 formalization (scoped).** Full machine-checked formalization of the PCL* convergence theorem requires formalizing optimal transport (Kantorovich metrics on distribution spaces), measure theory over function spaces, coalgebraic bisimulation, and PAC learning theory—none of which are currently available in Mathlib. We estimate this effort at 80–120K lines of Lean 4, representing a multi-year team effort. *We do not claim formal verification of the core theory.* As a pragmatic bridge, we provide 38 property-based tests (using `proptest`) that validate the core mathematical invariants computationally: bisimulation metric properties, PCL* convergence bounds, CoalCEGAR monotonicity and Galois distortion, PAC error decomposition, classifier error propagation, and differential testing against known automata. A feasible initial Lean 4 target would be the observation table closure/consistency invariants and the union bound composition, which require only basic real analysis from Mathlib ($\sim$5K LoC).
2. **Real LLM API validation.** The current Phase 0 results use stochastic mock LLMs. The critical open question is whether deployed LLMs (GPT-4, Claude, Llama) exhibit sufficiently stationary behavior within an audit window to admit tractable finite-state abstraction. We plan to deploy Caber against real APIs with CUSUM-based drift monitoring.

3. **Query efficiency.** Current query costs ($200–$500 per audit) can be reduced via: (a) active learning to prioritize informative queries, (b) caching responses across audits, and (c) incremental automaton updates when only a subset of behaviors change. We estimate a 5–10× reduction is achievable.

Availability. CABER is released under the MIT license. The repository includes all source code, mock LLMs, Phase 0 experiment scripts, classifier robustness analysis, and documentation for reproducing the experiments in this paper.

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A Proof Details

A.1 PCL* Convergence (Theorem 4.1)

Proof sketch. The convergence argument proceeds in three steps.

Step 1: Statistical membership queries. For each table entry $T(s, e)$, we draw m i.i.d. samples from the model's output distribution $\mathfrak{M}(s \cdot e)$. By Hoeffding's inequality on $[0, 1]$-bounded random variables,

$$\Pr\big[|T(s, e) - \mathfrak{M}(s \cdot e)| \geq t\big] \;\leq\; 2\exp(-2m\,t^2).$$

Setting $t = \varepsilon/(3n_0|E|)$ and applying a union bound over $n_0 \cdot |E|$ table entries gives the per-entry accuracy guarantee with confidence $\delta/3$. Note that we use Hoeffding's inequality (not a sub-Gaussian assumption), which applies because the output distribution is bounded in $[0, 1]$ by construction of the abstraction layer.

Step 2: PAC equivalence queries. Each equivalence query draws N_{eq} random prompts and compares the hypothesis H's predictions with the model's empirical distributions. By a union bound over the N_{eq} test prompts, the equivalence query has false-positive rate $\leq \delta/3$. The number of counterexamples before convergence is bounded by n_0 (the number of states), because each counterexample adds at least one new equivalence class to the observation table.

Step 3: Functor bandwidth bound. The key insight is that the effective alphabet size is governed by the covering number $N_\varepsilon(\text{Im}(\gamma_\alpha), d_K)$ rather than the raw alphabet size $|\Sigma_\alpha|$. This yields a total query complexity of $\tilde{\mathcal{O}}(\beta(F, \alpha) \cdot n_0 \cdot \log(1/\delta))$, where $\beta = \log N_\varepsilon$.

Combining the three steps via union bound: the total failure probability is $\delta_{\text{learn}} + \delta_{\text{eq}} \leq 2\delta/3 < \delta$, giving the stated guarantee. □

A.2 Classifier Error Propagation (Equation 1)

Proof. Let $\rho \in [0, 1)$ be the classifier error rate, i.e., each output label is independently corrupted with probability ρ. The corrupted observation table entry $\tilde{T}(s, e)$ satisfies

$$d_{\text{TV}}\big(\tilde{T}(s, e),\ T(s, e)\big) \ \leq\ \rho,$$

because the error acts as a noise channel that shifts the empirical distribution by at most ρ in total variation.

Substituting into the Hoeffding bound from Step 1:

$$\Pr\big[|\tilde{T}(s, e) - \mathfrak{M}(s \cdot e)| \geq t + \rho\big] \ \leq\ 2\exp(-2m\,t^2).$$

To achieve the same effective tolerance t as in the error-free case, we need the per-entry tolerance to absorb the ρ shift, yielding $\varepsilon_{\text{learn}}(\rho) = \varepsilon_{\text{learn}} + \rho$. More precisely, the learning error becomes $\varepsilon_{\text{learn}}/(1 - \rho)$ when accounting for the reduced effective sample quality.

The sample complexity adjusts as $m(\rho) = m_0/(1 - \rho)^2$ to maintain the same confidence level, since each sample has only a $(1 - \rho)$ probability of being correctly classified.

Combining: $\varepsilon_{\text{total}} \leq \varepsilon_{\text{learn}}/(1 - \rho) + \varepsilon_{\text{mc}} + \rho$. □

A.3 CoalCEGAR Monotonicity (Proposition 1)

Proof sketch. Let $\alpha \preceq \alpha'$ in the abstraction lattice. The Galois connection between the corresponding coalgebras ensures that for any safety property φ (a greatest fixed point):

$$\varphi_\alpha \ \leq\ \varphi_{\alpha'} + \delta(\alpha, \alpha'),$$

where $\delta(\alpha, \alpha') = \max_s d_{\text{TV}}(\gamma_\alpha(s), \gamma_{\alpha'}(s))$ is the Galois distortion. For safety properties (greatest fixed points), the abstraction can only under-approximate the satisfaction degree, so refinement can only increase it.

For liveness properties (least fixed points), the direction reverses: abstraction can over-approximate reachability, so refinement may *decrease* the satisfaction degree. The monotonicity guarantee for liveness requires $\delta(\alpha, \alpha') < \varepsilon_\varphi$, where ε_φ is the graded satisfaction margin. □